

Potato Blight Classification

By Group 1



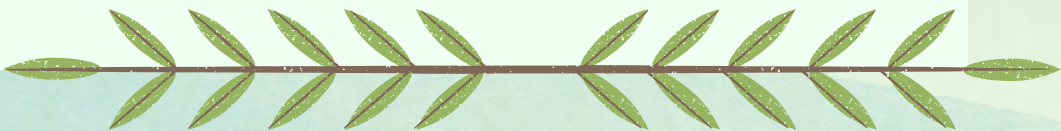


Introduction

Potato crops are highly susceptible to **Early Blight** (*Alternaria solani*) and **Late Blight** (*Phytophthora infestans*).

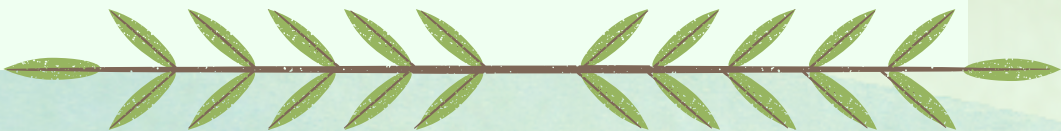
Phytophthora infestans literally translates to
'plant destroyer'

In fact, it was the culprit behind the Great Irish Famine (1845-1852) and continues to cause billions of crop losses annually.





Threat to Food Security / Livelihoods

- Many smallholder farmers depend on potato yields for both subsistence and income.
 - Left undetected, these diseases can devastate 30 - 70% of a crop, causing severe food insecurity and financial loss.
 - Traditional disease identification relies on experts conducting in-person field inspections - a resource that most smallholder farmers may not access in time. By the point visible symptoms have spread, the window for effective treatment is often already closed.
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The Solution

- This project employs *deep learning* to build an **automated potato blight classification system** using the PlantVillage dataset.
 - A user uploads an image of a potato leaf and in less than 2 seconds, the model classifies it into one of three categories (Early Blight, Late Blight, and Healthy)
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Stakeholders

- Smallholder Farmers: Early, accurate disease diagnosis to protect yield and income
 - Agricultural Officers: A scalable tool to assist field services
 - NGOs & Food Security Bodies: Reduction of crop loss in vulnerable communities
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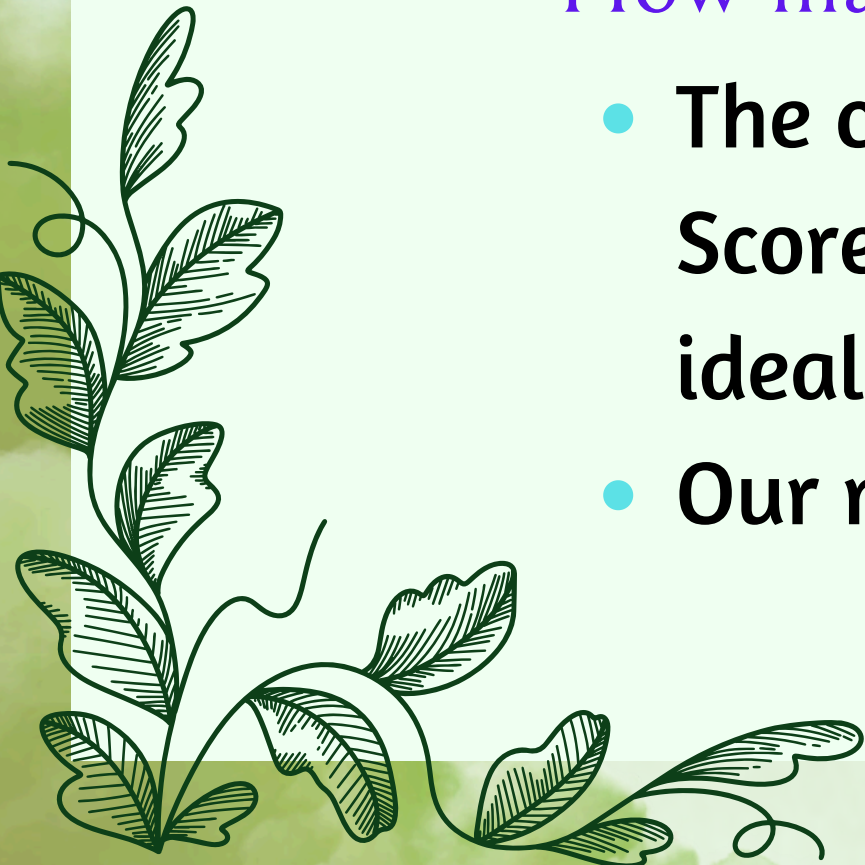
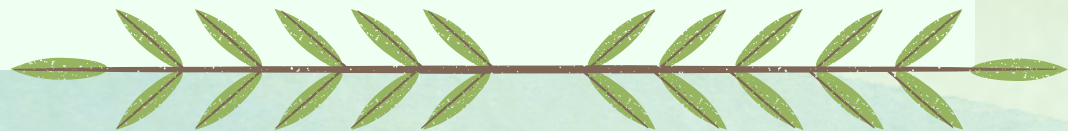


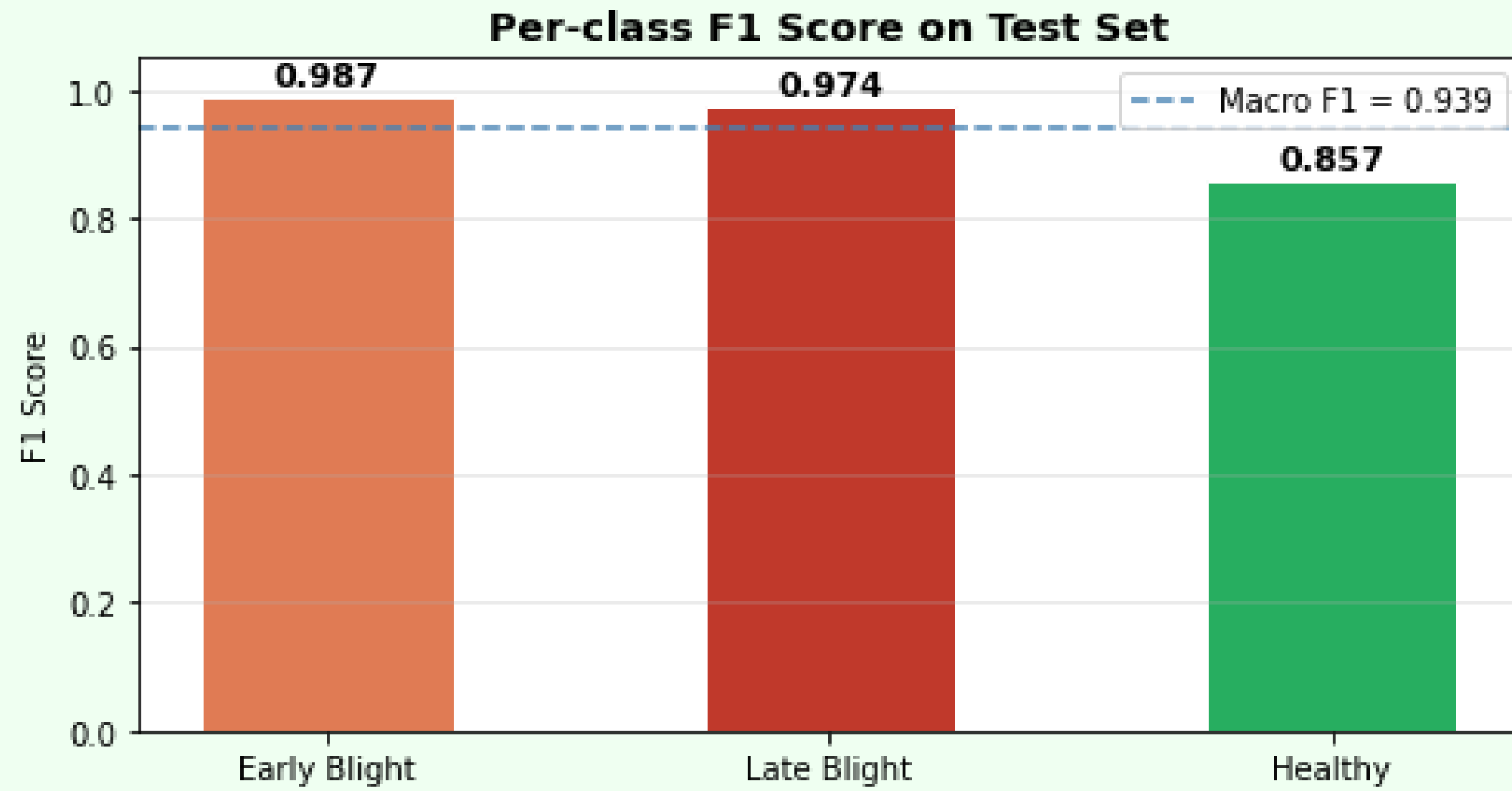
Data Handling

- We used a 2,152-image subset of the PlantVillage dataset (1000 Early Blight, 1000 Late Blight, 152 Healthy)
 - We addressed the imbalance using class weights. Images were normalized via resizing and scaling, and data augmentation was applied to the training set.
 - As for modeling, we evaluated two baseline models before utilizing transfer learning for improved performance and efficiency.
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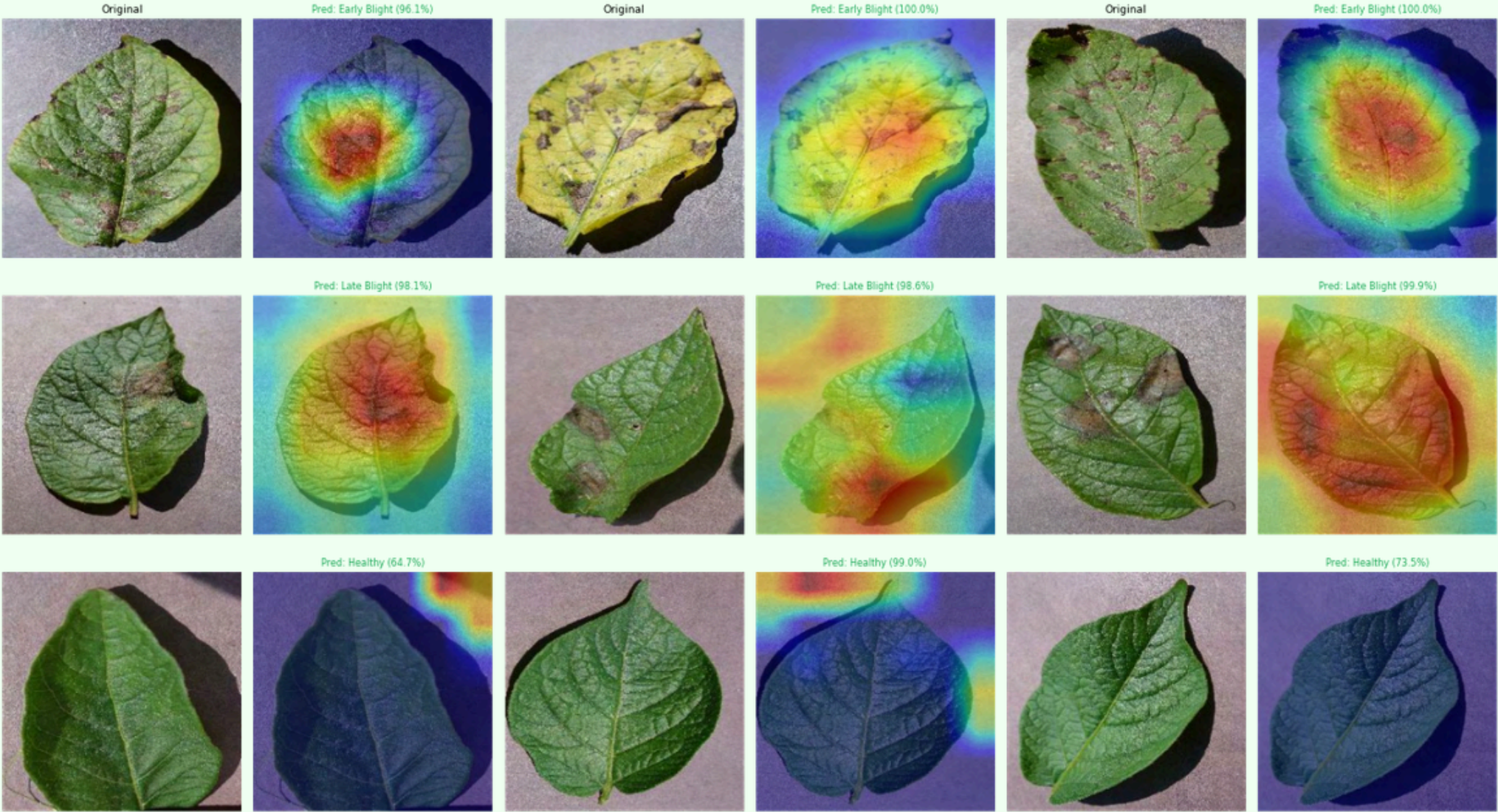


Reliability / Performance of the Model

- We need our model to score highly both in terms of “How many of the class predictions are actually that class?” & “How many of the actual class did we get?”
 - The core metric that answers those questions together is F1 Score, which ranges from 0 - 1, with a higher score being ideal.
 - Our model had a macro F1-Score of >0.9
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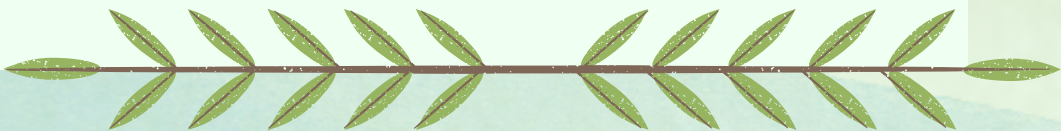


Grad-CAM - Random Samples





Practical Constraints

- The model was trained and evaluated exclusively on PlantVillage images - controlled, close-up, single-leaf photographs taken against uniform backgrounds under consistent lighting.
 - To get high-accuracy results, users must follow simple rules: *photograph a single leaf, close-up, ensure plain backgrounds, and use even light.*
 - The tool is explicitly trained on potato blight--it is not yet a general-purpose 'crop doctor' for other conditions.
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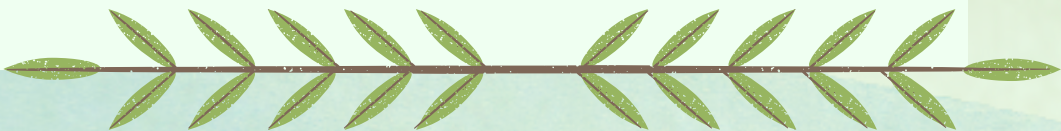
Next Steps

(Data Expansion): Partner with local agricultural groups to collect 1,000+ images of healthy leaves and real-world, noisy field photos to make the model blind to background clutter.



(Two-Stage AI): Upgrade software to automatically detect and crop a leaf out of a messy photo before analyzing it.

(Pilot Deployment): Launch a small-scale mobile pilot with a select group of agricultural officers to test usability.



Thank You

